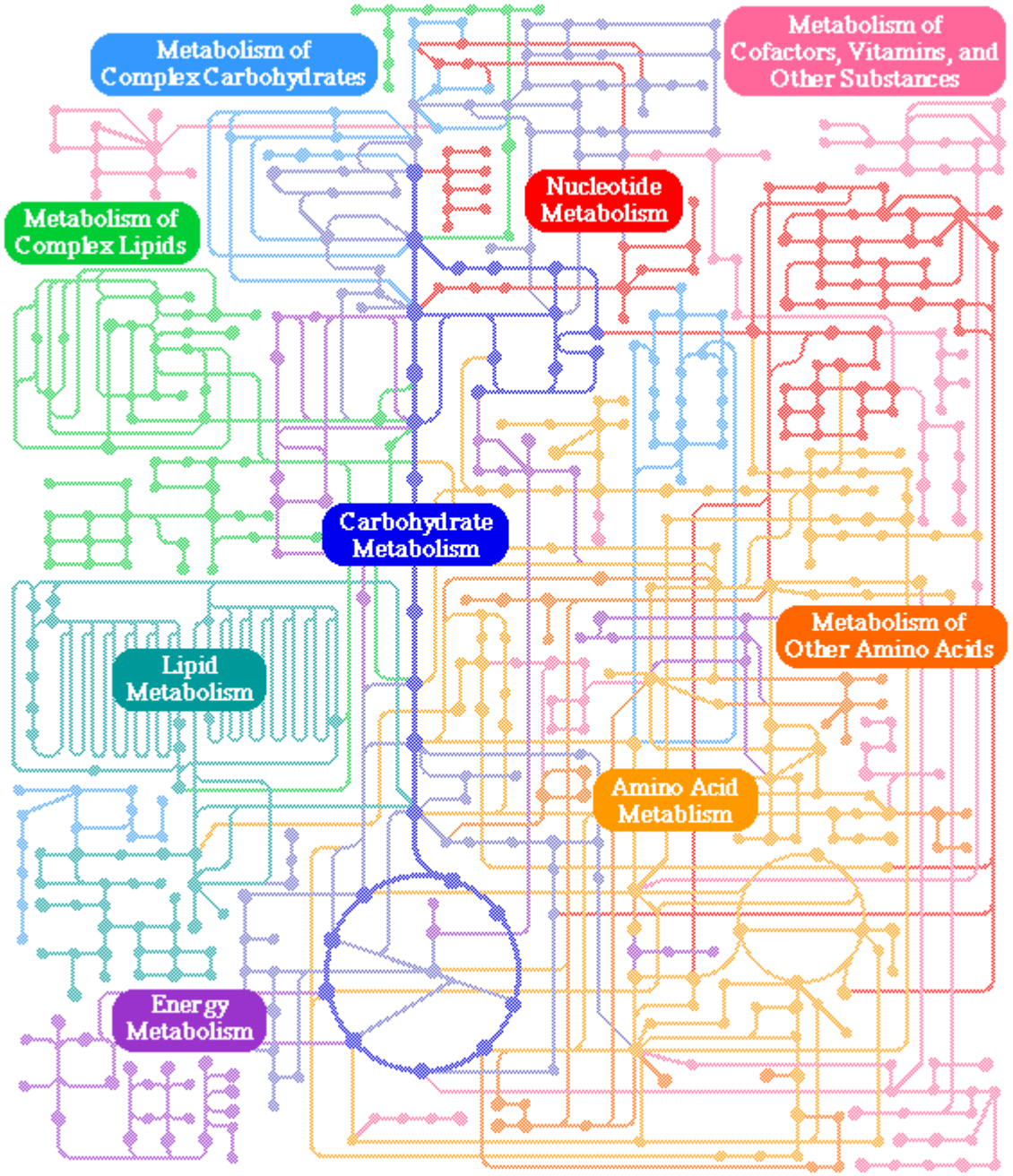
# ENZYMES

# WHY ENZYMES?

Accelerate and control the rates of vitally important biochemical reactions

- Greater reaction specificity
- Milder reaction conditions
- Capacity for regulation
- Enzymes are the agents of metabolic function.
- Metabolites have many potential pathways
- Enzymes make the desired one most favorable



# AN INTRODUCTION TO ENZYMES

- Life depends on powerful and specific catalysts: enzymes. Almost every biochemical reaction is catalyzed by an enzyme
- With the exception of a few catalytic RNAs, all known enzymes are proteins. Many require nonprotein coenzymes or cofactors for their catalytic function
- Enzymes are classified according to the type of reaction they catalyze. All enzymes have formal E.C. numbers and names, and most have trivial names
- Carbonic Anhydrase is the fastest enzyme. One molecule of carbonic anhydrase is capable of hydrating 106 CO2 molecules per second. Carbonic Anhydrase is an enzyme found in pancreatic cells, red blood cells, and gastric mucosa to name a few. It is involved in catalyzing a reversible reaction between water and carbon dioxide, resulting in the generation of carbonic acid

**TABLE 6–1** Some Inorganic Ions That Serve as Cofactors for Enzymes

| Ions | Enzymes |
|---|---|
| $Cu^{2+}$ | Cytochrome oxidase |
| $Fe^{2+}$ or $Fe^{3+}$ | Cytochrome oxidase, catalase, peroxidase |
| $K^{+}$ | Pyruvate kinase |
| $Mg^{2+}$ | Hexokinase, glucose 6-phosphatase, pyruvate kinase |
| $Mn^{2+}$ | Arginase, ribonucleotide reductase |
| Mo | Dinitrogenase |
| $Ni^{2+}$ | Urease |
| $Zn^{2+}$ | Carbonic anhydrase, alcohol dehydrogenase, carboxypeptidases A and B |

**TABLE 6–2** Some Coenzymes That Serve as Transient Carriers of Specific Atoms or Functional Groups

| Coenzyme | Examples of chemical groups transferred | Dietary precursor in mammals |
|---|---|---|
| Biocytin | $CO_2$ | Biotin |
| Coenzyme A | Acyl groups | Pantothenic acid and other compounds |
| 5′-Deoxyadenosylcobalamin (coenzyme $B_{12}$) | H atoms and alkyl groups | Vitamin $B_{12}$ |
| Flavin adenine dinucleotide | Electrons | Riboflavin (vitamin $B_2$) |
| Lipoate | Electrons and acyl groups | Not required in diet |
| Nicotinamide adenine dinucleotide | Hydride ion ($:H^{-}$) | Nicotinic acid (niacin) |
| Pyridoxal phosphate | Amino groups | Pyridoxine (vitamin $B_6$) |
| Tetrahydrofolate | One-carbon groups | Folate |
| Thiamine pyrophosphate | Aldehydes | Thiamine (vitamin $B_1$) |

# HOW ENZYMES WORK?

- Enzymes affect Reaction rates, Not Equilibria
- Enzymes enhance the rates by 5 to 17 orders of magnitude
- Enzymes enhance reaction by *lowering activation energy*
  - **Rearrangement of covalent bonds** during an enzyme-catalyzed reaction. Catalytic functional groups on an enzyme may form a transient covalent bond with a substrate and activate it for reaction, or a group may be transiently transferred from the substrate to the enzyme.
  - Non-covalent interactions between E & S stabilize the protein structure including hydrogen bonds and hydrophobic and ionic interactions. Formation of each weak interaction in the ES complex is accompanied by release of a small amount of free energy that stabilizes the interaction. The energy derived from enzyme-substrate interaction is called **binding energy**. Binding energy is a major source of free energy used by enzymes to lower the activation energies of reactions.
- The same binding energy that provides energy for catalysis also gives an enzyme its specificity

$$E + S \rightleftharpoons ES \rightleftharpoons EP \rightleftharpoons E + P \quad (6\text{–}1)$$

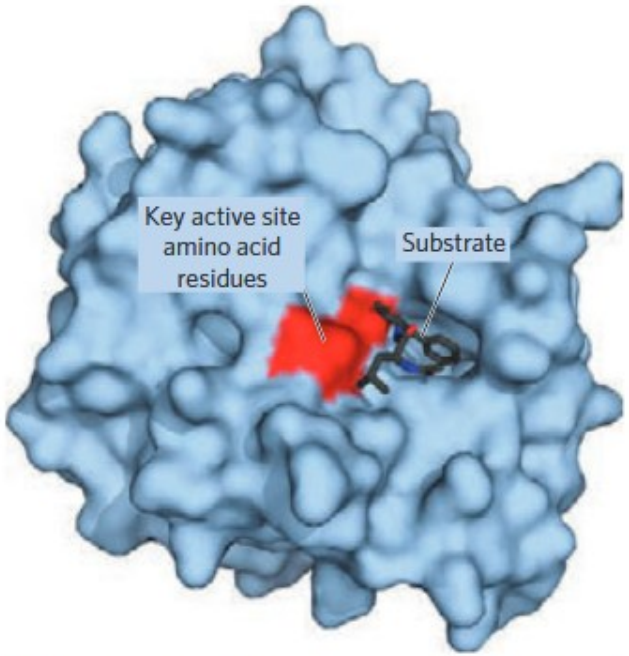


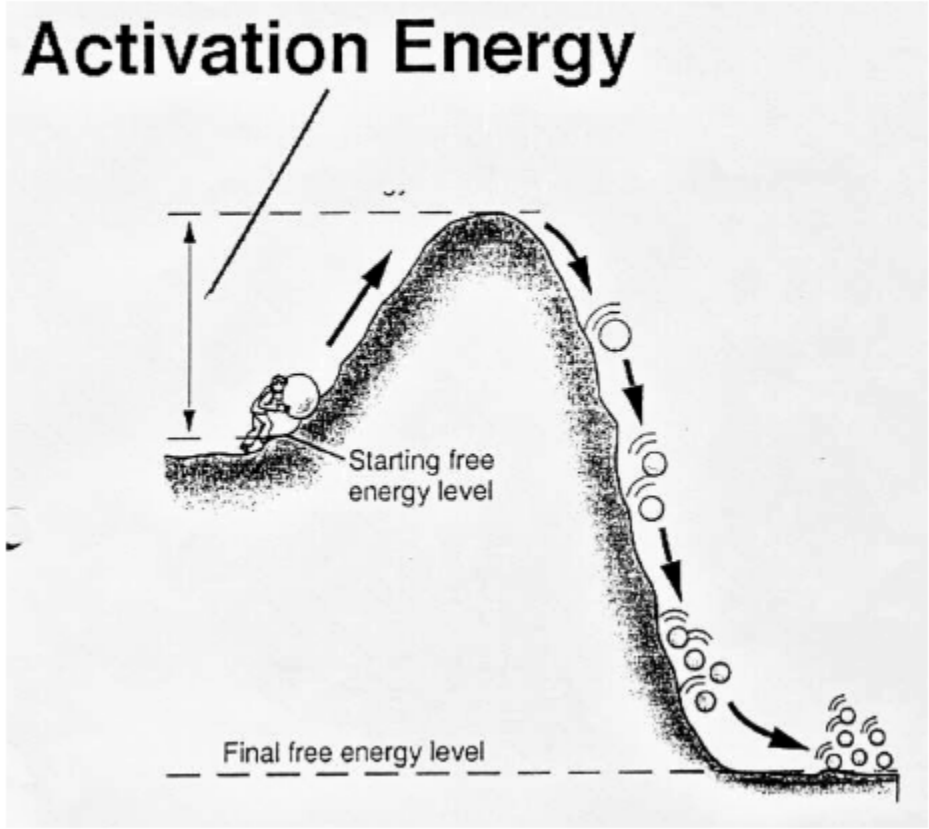

**FIGURE 6–1 Binding of a substrate to an enzyme at the active site.** The enzyme chymotrypsin with bound substrate (PDB ID 7GCH). Some key active-site amino acid residues appear as a red splotch on the enzyme surface.

| catalyst | rate enhancement |
|---|---|
| non-enzymatic (Pd) | $10^2$-$10^4$ fold |
| enzymatic | up to $10^{20}$ fold |

- **How much is $10^{20}$ fold?**

| catalyst | time for reaction |
|---|---|
| yes | 1 second |
| no | $3 \times 10^{12}$ years |

Supply energy by pushing a rock up a hill so that it can slide down the other side, releasing energy.




Think of activation energy as the BARRIER required to make a product.

Most stable product is the one with the lowest energy.

Most reactions require a "push" to get them started! "Push" is called "energy of activation" for reaction - Also represented by $E_A$

Reaction is thermodynamically favourable or SPONTANEOUS if $\Delta G^\circ$ is negative.
(Note: Spontaneous does NOT mean instantaneous; energy must be supplied to START a reaction which then proceeds with a release of energy)

4. i.e. Enzymes DO NOT change thermodynamics (can't make a reaction spontaneous)

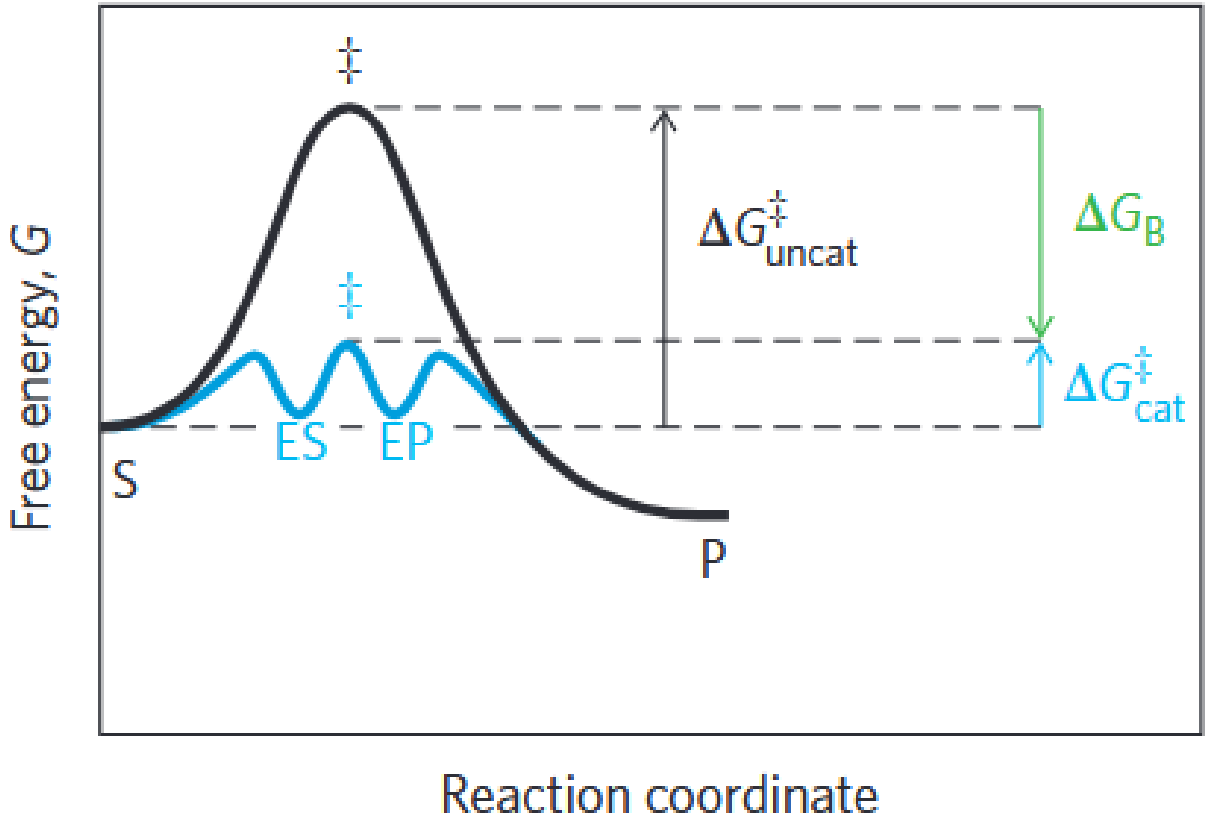
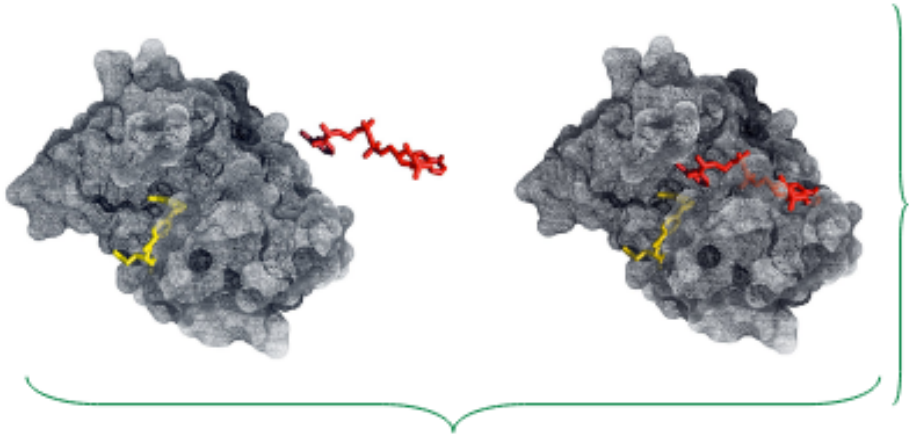
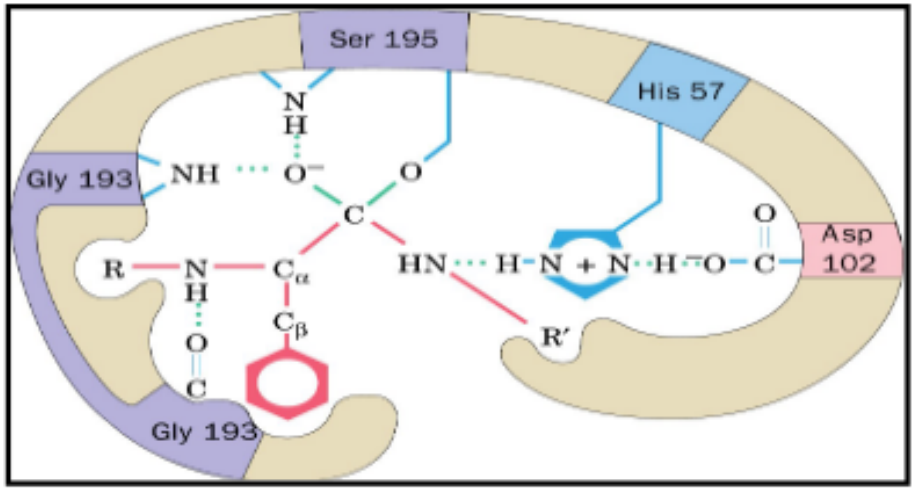



**FIGURE 6–6 Role of binding energy in catalysis.** To lower the activation energy for a reaction, the system must acquire an amount of energy equivalent to the amount by which $\Delta G^{\ddagger}$ is lowered. Much of this energy comes from binding energy ($\Delta G_B$) contributed by formation of weak noncovalent interactions between substrate and enzyme in the transition state. The role of $\Delta G_B$ is analogous to that of $\Delta G_M$ in Figure 6–5.

# Weak Interactions Between Enzyme and Substrate are Optimized in the Transition State






Non-Covalent Interactions!

Binding energy ($\Delta G_B$)---free energy released in forming multiple weak bonds and interactions between an enzyme and its substrate

Ensures favorable formation of the ES complex, but not too favorable

Binding energy is used to lower the activation energy barrier

Transition State:

a. Old bonds break and new ones form.

b. Substance is neither substrate nor product

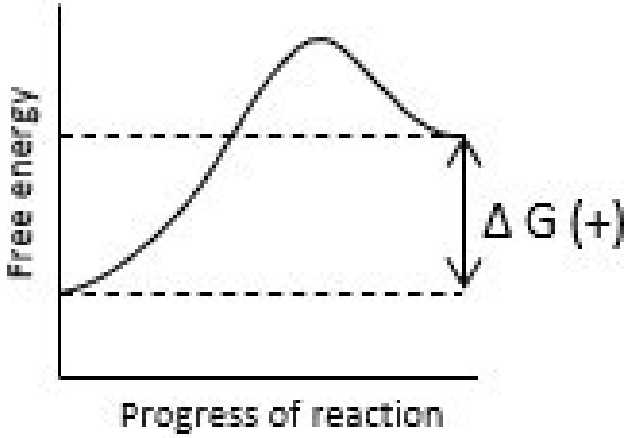
c. Unstable short lived species with an equal probability of going forward or backward.

d. Strained intermediate

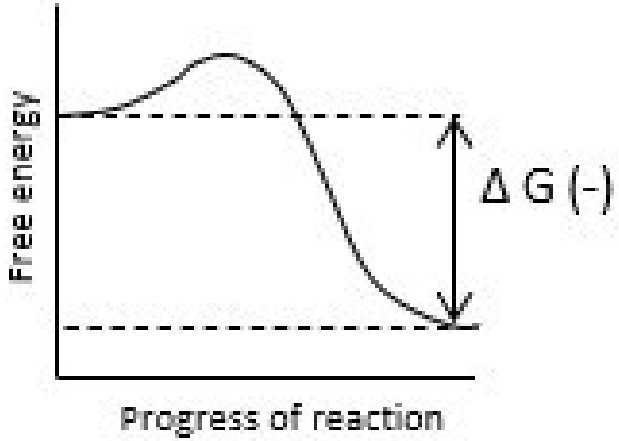
What type of interaction would occur between each group present on a substrate molecule and a functional group of the active site in an enzyme?

COOH
NH3+
OH
CH(CH3)2

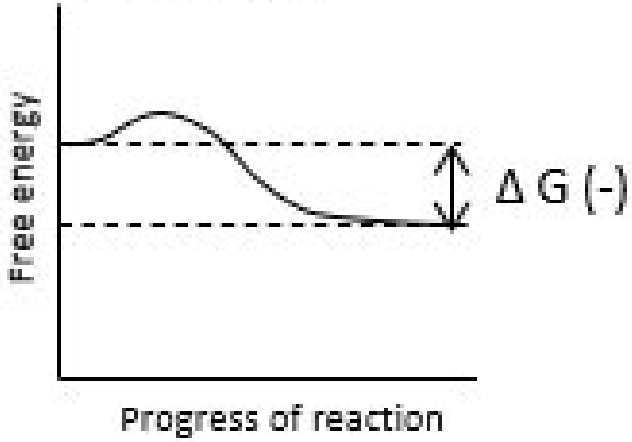
**(a)** Glucose + $P_i$ → Glucose 6-phosphate
(endergonic)




**(b)** ATP → ADP + $P_i$
(exergonic)




**(c)** Glucose + ATP → Glucose 6-phosphate + ADP
(exergonic)



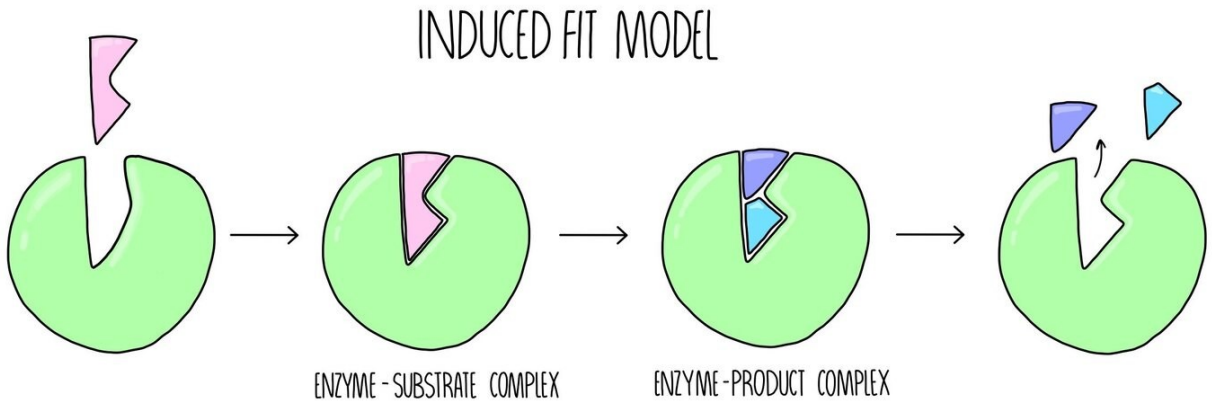
# Induced Fit Model

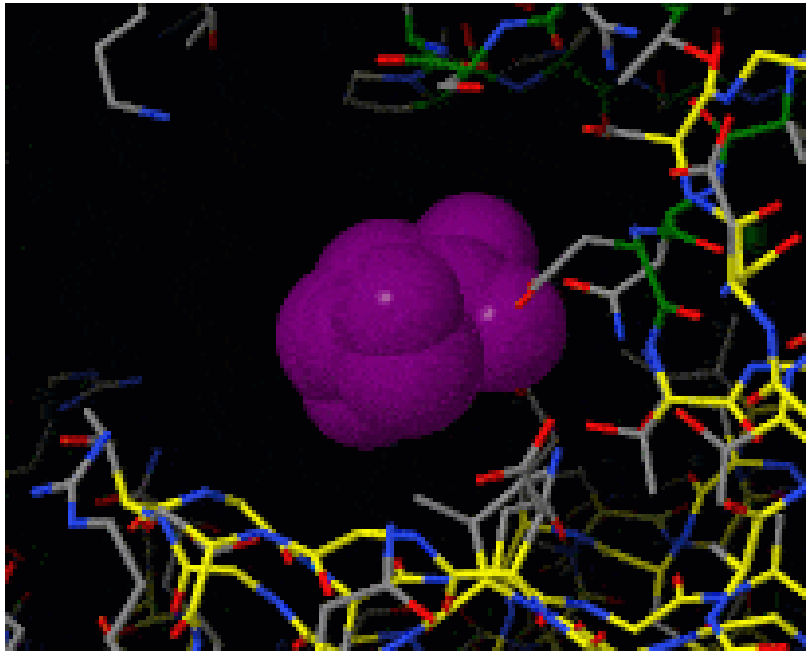
A. Takes into account the flexibility of proteins

B. A substrate fits into a general shape in the enzyme, causing the enzyme to change shape (conformation); close but not perfect fit of E + S

C. Change in protein configuration leads to a near perfect fit of substrate with enzyme

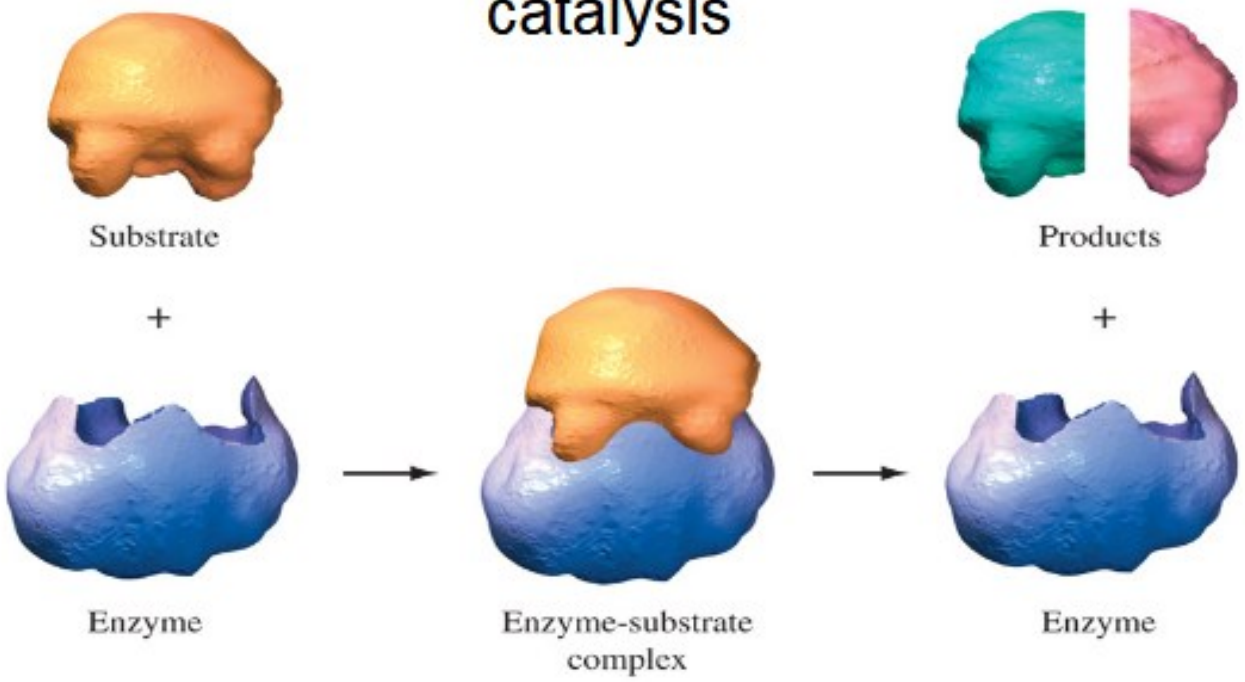
Refer to this video (Hexokinase in action) https://www.youtube.com/watch?v=KOyPakHIkaM



# Enzyme Catalysis

***Enzymes*** are proteins that provide a 3D surface for catalysis

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$$E + S \underset{}{\overset{\text{Step I}}{\rightleftharpoons}} ES \overset{\text{Step II}}{\rightleftharpoons} ES^* \overset{\text{Step III}}{\rightleftharpoons} EP \overset{\text{Step IV}}{\rightleftharpoons} E + P$$

| | | | | |
|---|---|---|---|---|
| Enzyme + substrate | Enzyme–substrate complex | Transition state | Enzyme–product complex | Enzyme + product |

# Classification of Enzymes

**TABLE 6–3** International Classification of Enzymes

| Class no. | Class name | Type of reaction catalyzed |
|---|---|---|
| 1 | Oxidoreductases | Transfer of electrons (hydride ions or H atoms) |
| 2 | Transferases | Group transfer reactions |
| 3 | Hydrolases | Hydrolysis reactions (transfer of functional groups to water) |
| 4 | Lyases | Cleavage of C—C, C—O, C—N, or other bonds by elimination, leaving double bonds or rings, or addition of groups to double bonds |
| 5 | Isomerases | Transfer of groups within molecules to yield isomeric forms |
| 6 | Ligases | Formation of C—C, C—S, C—O, and C—N bonds by condensation reactions coupled to cleavage of ATP or similar cofactor |

https://proteopedia.org/wiki/index.php/Proteopedia:Table_of_Contents

*Which enzyme is given EC 1.1.1.1?*

# Protein structure

Scaffold to support and position active site

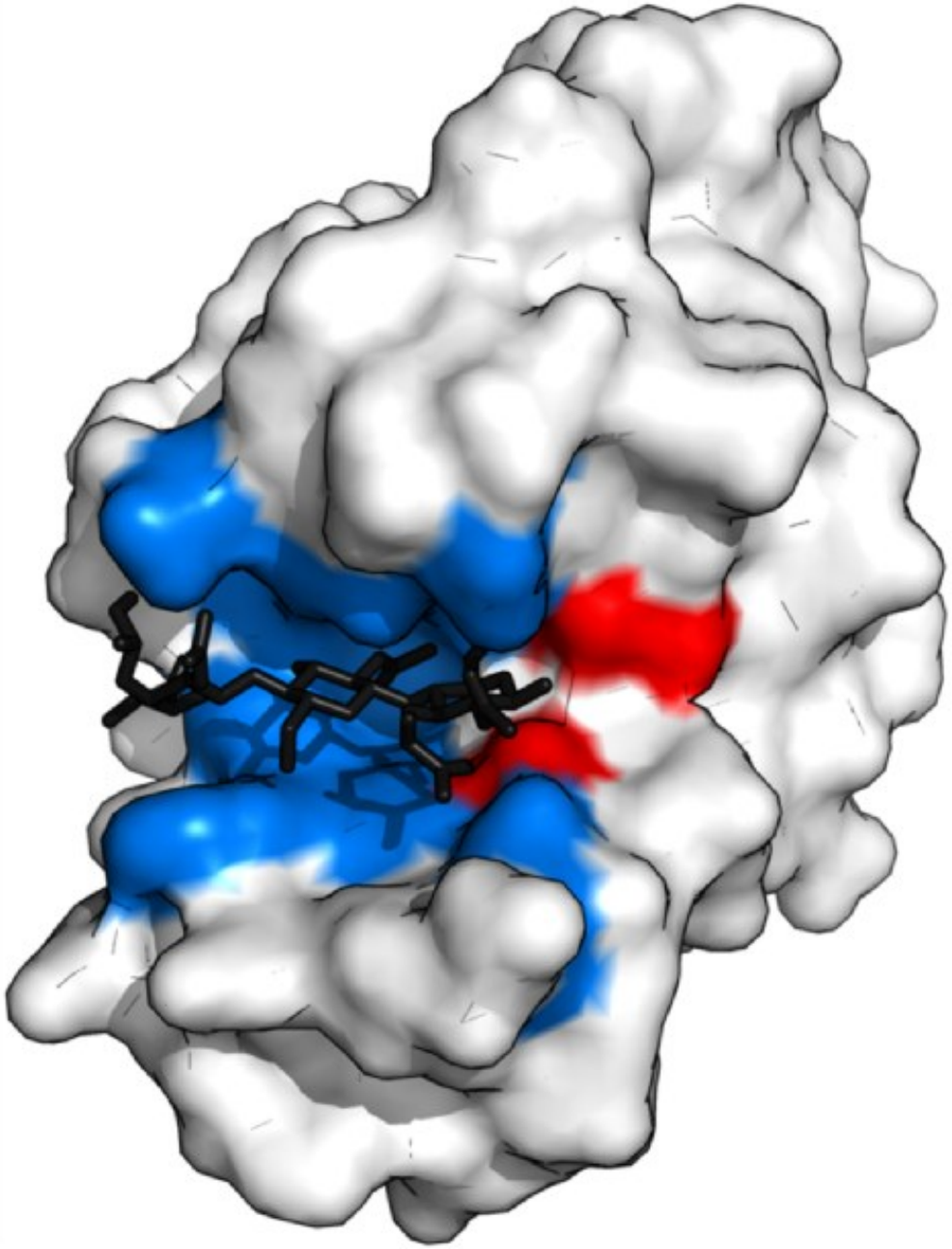
## Active site

### Binding sites

Bind and orient substrate(s)

### Catalytic site

Reduce chemical activation energy



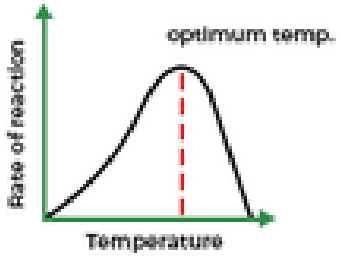
# ACTIVE SITE

- Pocket in the enzyme where substrates bind and catalytic reaction occurs
- Has SPECIFICITY – can discriminate among possible substrate molecules
  - Most recognize a functional group (Group specificity)
  - Some only recognize one type of molecule (e.g. D vs L isomers)
- Active site is a relatively small 3-D region within the enzyme
- Substrates bind in active site by weak non-covalent interactions
  - A. hydrogen bonding
  - B. hydrophobic interactions
  - C. ionic interactions
- It provides orientation and proximity for most effective catalysis.
- The ENERGY derived from these interactions = "BINDING ENERGY"
- Using this energy, enzymes both lower the activation energy and stabilize the transition state complex (ES*). Each non-covalent interaction provides energy to stabilize the transition state

# Factors affecting ENZYME activity

## Temperature

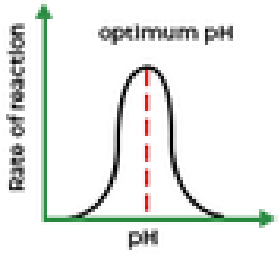
BIG influence
More heat = More kinetic energy




But if too high
enzyme is denatured

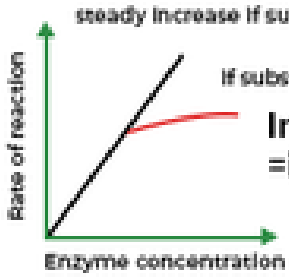
## pH

Enzymes have optimum pH




If higher/lower
H+ in acid / OH- in alkaline
can interfere enzyme structure

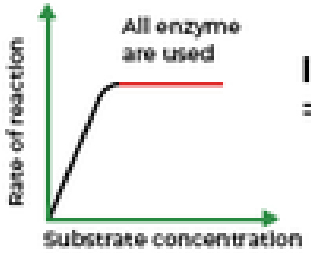
## Enzyme concentration




Increase enzyme concentration
=increase rate of reaction

**Until substrates amount are limited**

## Substrate concentration

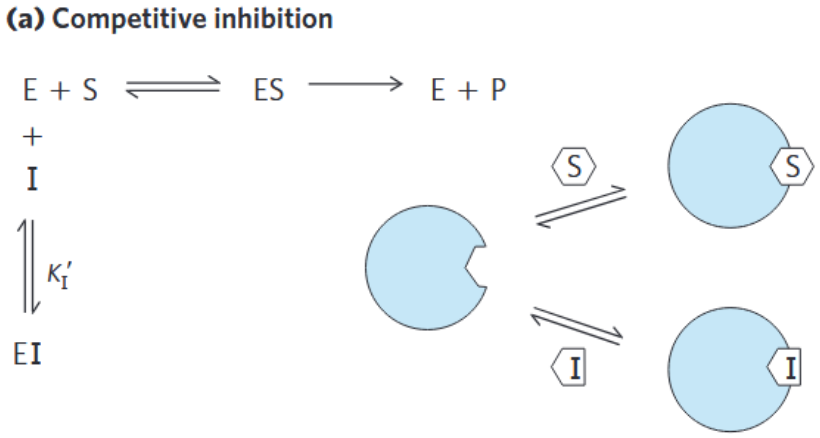


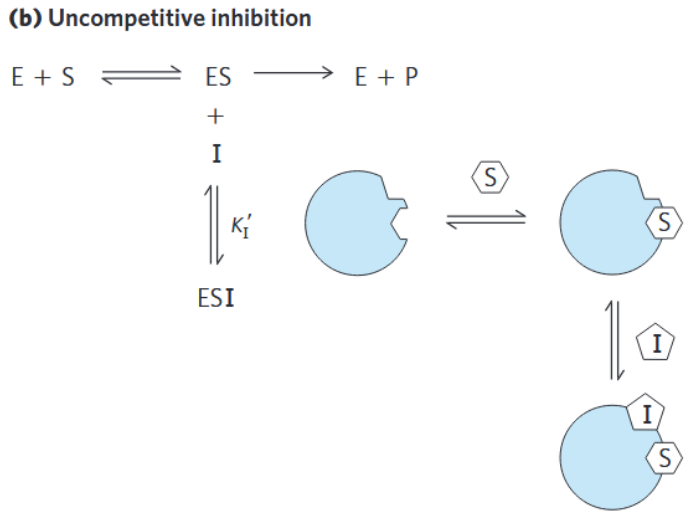

Increase substrate concentratio
=increase rate of reaction

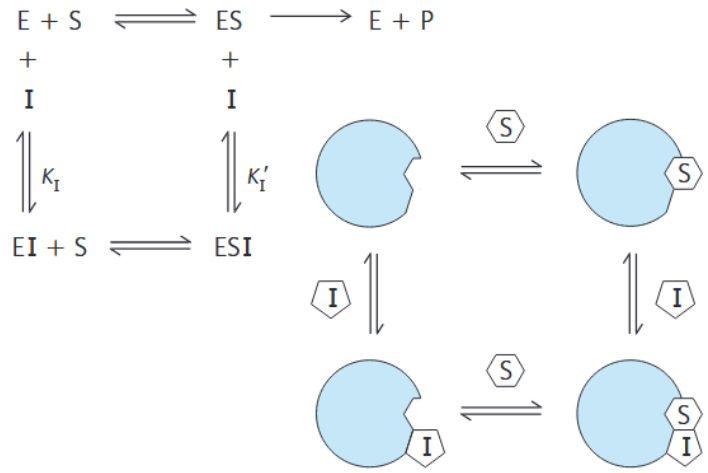
**Until active site of enzyme are used**

# Reversible Inhibition

| Competitive | Uncompetitive | Noncompetitive |
|---|---|---|
| 1. The inhibitor can combine with the free enzyme in such a way that it competes with the normal substrate for binding at the active site. It is also called as Substrate analogue Inhibition. | 1.The inhibitor does not combine with the free enzyme or affects its reaction with its normal substrate; however, it does combine with the enzyme-substrate complex. | 1. The inhibitor can combine with either the free enzyme or the enzyme-substrate complex, interfering with the action of both. Inhibitors bind to a site on the enzyme other than the active site. |
| 2. An enzyme-inhibitor complex is formed [EI], analogous to the enzyme-substrate complex [ES]. Higher substrate concentration can lower the rate of inhibition. | 2. An inactive enzyme- substrate-inhibitor complex [ESI] is formed here which cannot undergo further reaction to yield the normal product. | 2. Inhibitors often to deform the enzyme, so that these do not form the [ES] complex at its normal rate and once formed, the [ES] complex does not decompose at the normal rate to yield products. Two inactive complexs, [ESI] & [EI] are formed. |
| 3. The degree of inhibition depends on the relative concentrations of the substrate and the inhibitor. | 3. The degree of inhibition may increase when the substrate concentration is increased. | 3. The degree of inhibition is not reversed by increasing the substrate concentration. |
| 4. Example :<br>Enzyme – Succinate Dehydrogenase<br>Substrate – Succinate<br>Inhibitors – Malonate,Glutarate, Oxalate etc. | 4. Example :<br>Rare in one-substrate reaction; but common in Bi-substrate reaction. | 4. Example :<br>Metal ion-requiring enzymes can be inhibited by chelating agents like EDTA. |
| 5. Kinetics :<br>Slope [$K_m/V_{max}$] is changed; Intercept on ordinate [$1/V_{max}$] is not changed. | 5. Kinetics :<br>Slope [$K_m/V_{max}$] is not changed; Intercept on ordinate [$1/V_{max}$] is changed. | 5. Kinetics :<br>Slope [$K_m/V_{max}$] is changed; Intercept on ordinate [$1/V_{max}$] is also changed. |



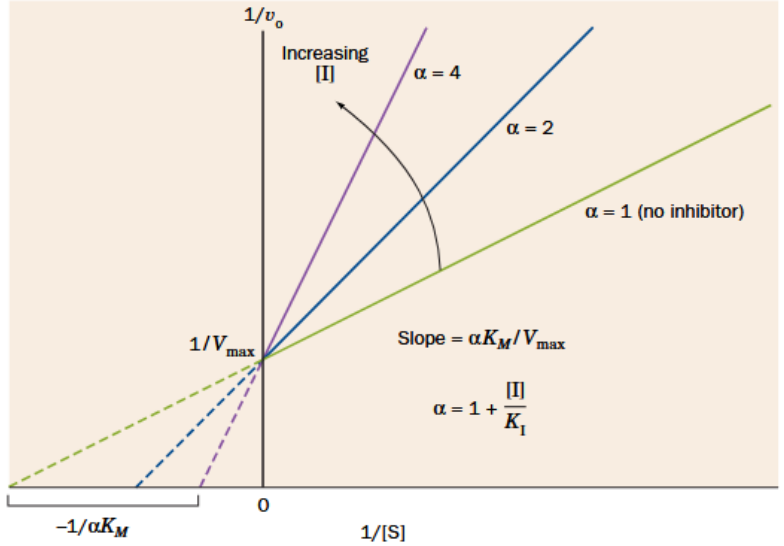




**FIG. 12-8 A Lineweaver–Burk plot of the competitively inhibited Michaelis–Menten enzyme described by Fig. 12-7.** Note that all lines intersect on the $1/v_o$ axis at $1/V_{max}$. The varying slopes indicate the effect of the inhibitor on $\alpha K_M = K_M^{app}$. **WPLS See the Animated Figures.**

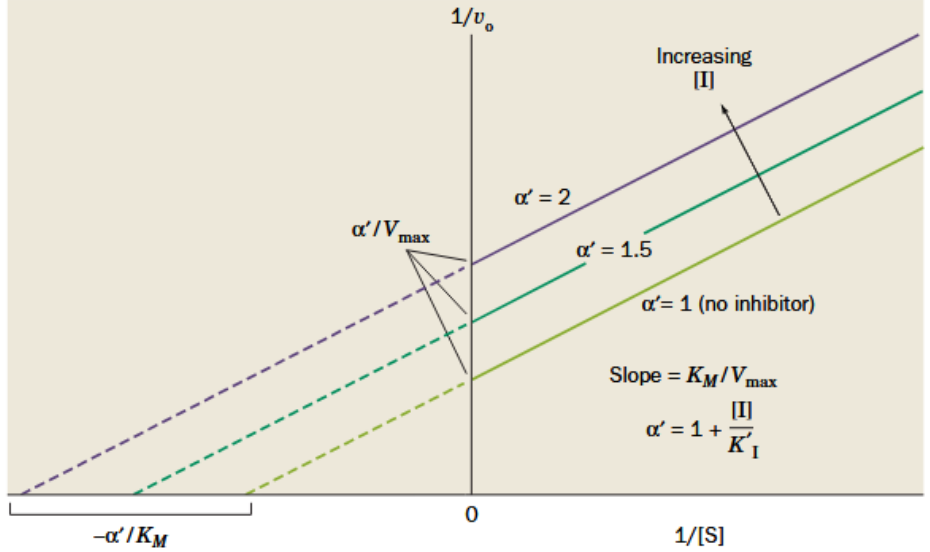



**FIG. 12-9 A Lineweaver–Burk plot of a Michaelis–Menten enzyme in the presence of an uncompetitive inhibitor.** Note that all lines have identical slopes of $K_M/V_{max}$. **WPLS See the Animated Figures.**

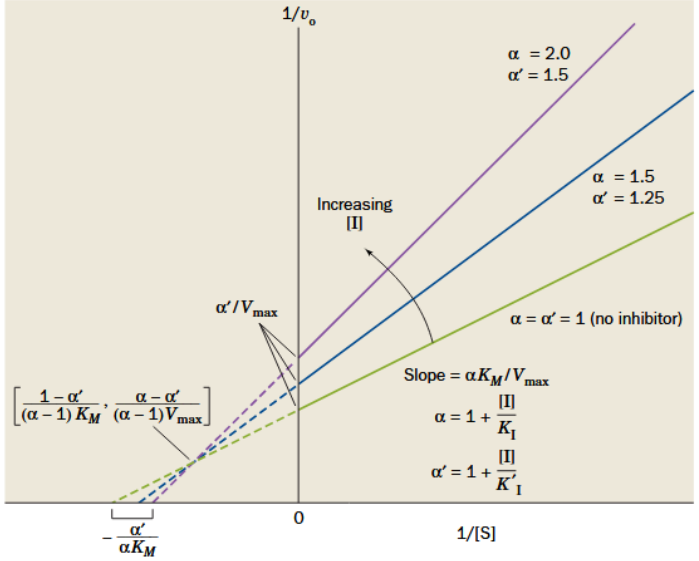



**FIG. 12-10 A Lineweaver–Burk plot of a Michaelis–Menten enzyme in the presence of a mixed inhibitor.** Note that the lines all intersect to the left of the $1/v_o$ axis. The coordinates of the intersection point are given in brackets. When $K_I = K_I'$ ($\alpha = \alpha'$), the lines intersect on the 1/[S] axis at $-1/K_M$. **WPLS See the Animated Figures.**

| Type of enzyme inhibitor | Enzyme inhibitor (drug) | Enzyme Target | Pharmaceutical use |
|---|---|---|---|
| Competitive reversible inhibitors | Lovastatin, Pravastatin, other statins | HMG-CoA reductase | Cholesterol lowering |
| | Captopril, enalapril | Angiotensin converting enzyme | Hypertension |
| | Saquinavir, indinavir, ritonavir | HIV protease | AIDS |
| | Acetazolamide | Carbonic anhydrase | Glaucoma |
| | Viagra, Levitra | Phosphodiesterase | Erectile dysfunction |
| | Gleevec | Bcr-Abl kinase | Cancer |
| | Methotrexate | Dihydrofolate reductase | Cancer, bacterial infection |
| | Nevirapine, efavirenz | HIV reverse transcriptase | AIDS |
| | SB-715992 | KSP kinesin | Cancer |
| | PD0325901, CI-1040 | MAP kinase kinase (MEK) | Cancer |
| Non competitive reversible inhibitors | TF-505 | Steroid 5a-reductase | Benign prostate Hyperplasia |
| | Etoposide | Topoisomerase II | Cancer |
| | Tacrine | Acetylcholinesterase | Alzheimer's disease |
| | Trazodone | Adenosine deaminase | Depression |
| | Mycophenolic acid,VX-148 | Inosine 5'-monophosphate dehydrogenase | Immunosuppression, cancer |
| | Finasteride, epristeride, dutasteride | Steroid 5α-reductase | Benign prostate hyperplasia, male pattern baldness |
| Uncompetitive reversible inhibitors | Valproic acid | UDP-glucuronosyltransferases | Xenobiotic metabolism |
| | Camptothecin | Topoisomerase I | Cancer |
| | Ciglitazone | 15-Hydroxyprostaglandin, Dehydrogenase | Inflammatory diseases |

# ENZYMES AND COFACTORS

Enzymes are composed of one or several polypeptide chains. However, in many cases, non-protein constituents called cofactors are bound to the enzyme to make it catalytically active. In these instances, the protein portion of the enzyme is called the apoenzyme.

Cofactor: An organic molecule or ion (usually a metal ion) that is required by an enzyme for its activity. It may be attached either loosely (coenzyme) or tightly (prosthetic group).

Three kinds of cofactors may be identified: prosthetic groups, coenzymes, and metal ions.

- Coenzyme: A low-molecular-weight, non-protein organic compound participating in enzymatic reactions as a dissociable acceptor or donor of chemical groups or electrons.
- Prosthetic Group: A tightly bound, specific nonpolypeptide unit in a protein determining and involved in its biological activity.
- Metal Ions: number of enzymes require metal ions for their activity which form coordination bonds with side chains at the active site and at the same time form one or more cordination bonds with the substrate, e.g., zinc is a cofactor for the proteolytic enzyme carboxypeptidase.